Angles & Parallel Lines Maze

Name _____ Period ____

Start in the top-left room. Solve the problem, find the door with your answer, and go through it. Show your work on your answer grid.

START

Vertical angles. Find x.

$(3x + 10)°$ and $70°$

$x = 20$

Find x.

Linear pair

$2x°$ and $(x + 30)°$

$x = 50$

Lines ∥. Find x.

Corresponding ∠s

$(5x - 15)°$, $(3x + 21)°$

$x = 3$

Lines ∥. Find the angle measure.

Alternate interior ∠s

$(9x - 20)°$, $(5x + 24)°$

$x = \frac{80}{3}$

$x = 60$

$x = 30$

$x = 70$

$x = 18$

$x = -18$

$11°$

Lines ∥. Find the angle measure.

Corresponding ∠s

$(8x - 2)°$, $(6x + 14)°$

$x = 41$

Lines ∥. Find x.

Same-side interior ∠s

$(2x + 20)°$, $(3x + 5)°$

$66°$

Lines ∥. Find the angle measure.

Alternate interior ∠s

$(4x + 6)°$, $(6x - 24)°$

$96°$

Find x.

Vertical angles

$(2x + 41)°$, $(4x - 5)°$

$113°$

$x = 31$

$x = 15$

$114°$

$15°$

$68°$

$x = -23$

Lines ∥. Find the angle measure.

Alternate exterior ∠s

$(7x - 8)°$, $(5x + 22)°$

$83°$

Find the angle measure.

Vertical angles

$(5x - 12)°$, $(3x + 16)°$

$58°$

Lines ∥. Find the larger ∠.

Same-side interior ∠s

$(6x + 4)°$, $(4x - 4)°$

$90°$

Lines ∥. Find x.

Same-side interior ∠s

$(x + 15)°$, $(2x + 21)°$

$x = 24$

$97°$

$120°$

$14°$

$112°$

$18°$

$x = -6$

Find x.

Linear pair

$(5x + 3)°$, $(3x + 9)°$

$27°$

Linear pair. Find m∠1.

$m\angle 1 = (4x + 12)°$

$m\angle 2 = (2x + 6)°$

$60°$

Lines ∥. Find x.

Same-side interior ∠s

$3(x + 10)°$, $2(x - 5)°$

$x = 32$

FINISH

You made it!

Angles & Parallel Lines Maze: Answer Grid

Name _____ Period ____

Write each answer in the order you found it, and show your work for that room.

1	Answer: _____	Work:
2	Answer: _____	Work:
3	Answer: _____	Work:
4	Answer: _____	Work:
5	Answer: _____	Work:
6	Answer: _____	Work:
7	Answer: _____	Work:
8	Answer: _____	Work:
9	Answer: _____	Work:
10	Answer: _____	Work:

Angles & Parallel Lines Maze ANSWER KEY

Start in the top-left room. Solve the problem, find the door with your answer, and go through it. Show your work on your answer grid.

START

1 Vertical angles. Find x.

$(3x + 10)°$ and $70°$

$x = 20$

2 Find x.

Linear pair

$2x°$ and $(x + 30)°$

$x = 50$

3 Lines ∥. Find x.

Corresponding ∠s

$(5x - 15)°$, $(3x + 21)°$

$x = 3$

Lines ∥. Find the angle measure.

Alternate interior ∠s

$(9x - 20)°$, $(5x + 24)°$

$x = \frac{80}{3}$

$x = 60$

$x = 30$

$x = 70$

$x = 18$

$x = -18$

11°

Lines ∥. Find the angle measure.

Corresponding ∠s

$(8x - 2)°$, $(6x + 14)°$

$x = 41$

5 Lines ∥. Find x.

Same-side interior ∠s

$(2x + 20)°$, $(3x + 5)°$

66°

4 Lines ∥. Find the angle measure.

Alternate interior ∠s

$(4x + 6)°$, $(6x - 24)°$

96°

Find x.

Vertical angles

$(2x + 41)°$, $(4x - 5)°$

113°

$x = 31$

$x = 15$

114°

15°

68°

$x = -23$

6 Lines ∥. Find the angle measure.

Alternate exterior ∠s

$(7x - 8)°$, $(5x + 22)°$

8 Find the angle measure.

Vertical angles

$(5x - 12)°$, $(3x + 16)°$

9 Lines ∥. Find the larger ∠.

Same-side interior ∠s

$(6x + 4)°$, $(4x - 4)°$

Lines ∥. Find x.

Same-side interior ∠s

$(x + 15)°$, $(2x + 21)°$

83°

58°

90°

$x = 24$

97°

120°

14°

112°

18°

$x = -6$

Find x.

Linear pair

$(5x + 3)°$, $(3x + 9)°$

7 Linear pair. Find m∠1.

$m\angle 1 = (4x + 12)°$

$m\angle 2 = (2x + 6)°$

10 Lines ∥. Find x.

Same-side interior ∠s

$3(x + 10)°$, $2(x - 5)°$

27°

60°

$x = 32$

FINISH

You made it!

Angles & Parallel Lines Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$x = 20$**	$3x + 10 = 70 \rightarrow 3x = 60 \rightarrow x = 20$
2	Answer: **$x = 50$**	$2x + x + 30 = 180 \rightarrow 3x = 150 \rightarrow x = 50$
3	Answer: **$x = 18$**	$5x - 15 = 3x + 21 \rightarrow 2x = 36 \rightarrow x = 18$
4	Answer: **66°**	$4x + 6 = 6x - 24 \rightarrow x = 15 \rightarrow 4(15) + 6 = 66$
5	Answer: **$x = 31$**	$5x + 25 = 180 \rightarrow 5x = 155 \rightarrow x = 31$
6	Answer: **97°**	$7x - 8 = 5x + 22 \rightarrow x = 15 \rightarrow 7(15) - 8 = 97$
7	Answer: **120°**	$6x + 18 = 180 \rightarrow x = 27 \rightarrow m\angle 1 = 4(27) + 12 = 120$
8	Answer: **58°**	$5x - 12 = 3x + 16 \rightarrow x = 14 \rightarrow 5(14) - 12 = 58$
9	Answer: **112°**	$10x = 180 \rightarrow x = 18 \rightarrow 6(18) + 4 = 112$
10	Answer: **$x = 32$**	$3x + 30 + 2x - 10 = 180 \rightarrow 5x + 20 = 180 \rightarrow x = 32$